

All three pipeline views on TEST — the magnitude residual collapses the  $a_{\text{vert}}$  outlier tail

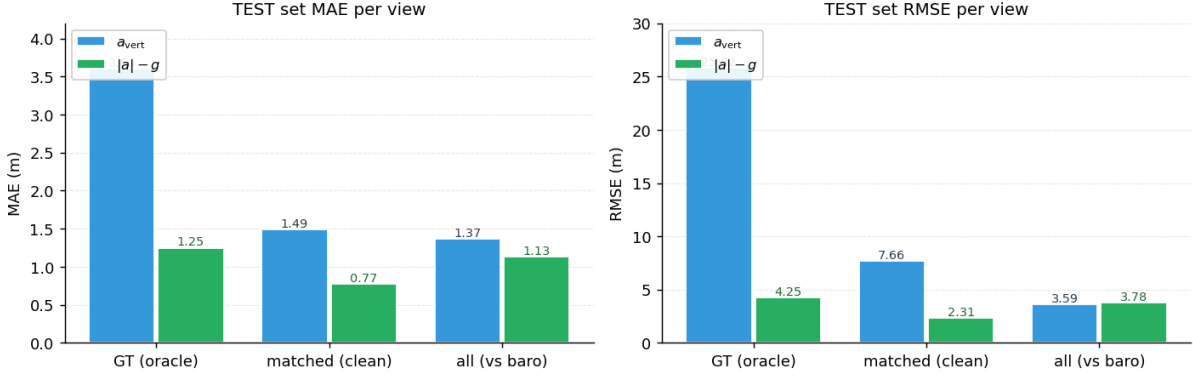


Figure 93: All three pipeline views on the TEST split. The GT-view RMSE of 25.82 m under  $a_{\text{vert}}$  is the most extreme outlier signature — a small number of TEST rides have catastrophic projection collapse that the matched-filter cannot fit. The same rides under  $|a| - g$  contribute single-digit-metre residuals (RMSE 4.25 m). The median-absolute-error column would be flat across all bars (it is  $\approx 0.29$  m in every cell of Table 29): the gap lives entirely in the tail.

| View                                  | $n$ | MAE    | Median | RMSE   | $\leq 0.5$ m | $\leq 1.5$ m |
|---------------------------------------|-----|--------|--------|--------|--------------|--------------|
| GT segments (predictor on every GT)   | 108 | 1.25 m | 0.29 m | 4.25 m | 70.4%        | 85.2%        |
| Matched segmenter (clean predictions) | 98  | 0.77 m | 0.26 m | 2.31 m | 68.4%        | 93.9%        |
| All segments vs barometer             | 100 | 1.13 m | 0.31 m | 3.78 m | 67.0%        | 90.0%        |

Table 32: Held-out test split under  $|a| - g$ . The numbers to watch: matched-view MAE 0.77 m versus the baseline  $a_{\text{vert}}$  section’s  $\sim 1.49$  m on the same data (§13).

### 13.3 MAE at a glance

### 13.4 Error CDFs

### 13.5 Signed-error PDFs and scatter

### 13.6 False-positive characterisation

### 13.7 Per-experiment MAE

### 13.8 Accepted-only (post quality-filter)

The trapezoid predictor’s quality filter is unchanged from the baseline run — only the input signal moved. The same set of features (joint  $R^2$ , cruise-velocity CV, anchor ratio, out-of-lobe residual density, pre-/post-gravity angle) classifies each prediction; the accept-rate under  $|a| - g$  is 96.8% on clean segments and 0.0% on FP segments.

### 13.9 Summary vs. the $a_{\text{vert}}$ baseline

The headline numbers under  $|a| - g$  in this section can be read side-by-side against the baseline reported in §13. The pattern established by §12 holds at the pipeline level:

- **Test-split matched MAE collapses** from  $\sim 1.49$  m to 0.77 m. The catastrophic outliers that drove the baseline’s TEST RMSE toward double digits do not appear under  $|a| - g$ .
- **False positives drop sharply:** the magnitude residual does not afford the stale- $\hat{g}$  projection path that turned horizontal user motion into phantom trapezoids. 0 FP segments

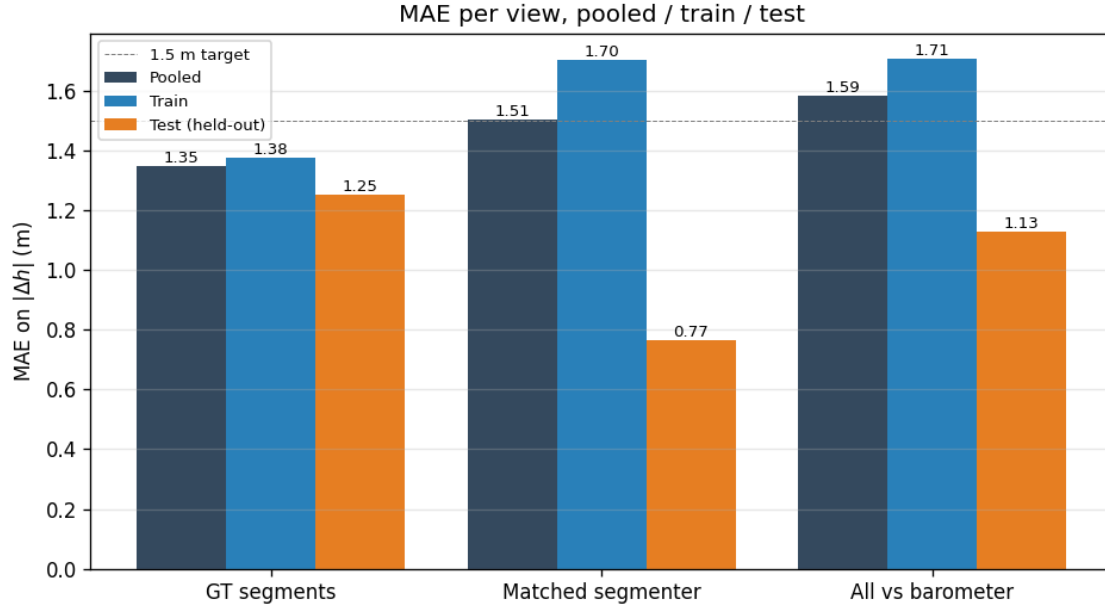


Figure 94: Pooled / train / test MAE per view,  $|a| - g$  run. Methodology mirrors Figure on page 107; the bars here can be compared cell-by-cell against the baseline figure. Headline: TEST matched-view MAE is 0.77 m, RMSE 2.31 m.

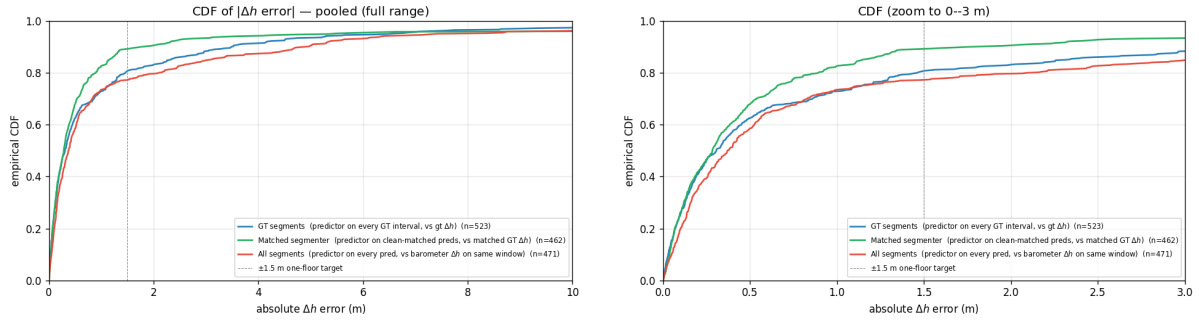


Figure 95: Pooled CDF of  $|\Delta h_{\text{pred}} - \Delta h_{\text{truth}}|$  under  $|a| - g$ . Left: full range. Right: zoom to 0–3 m, where most of the deployment-relevant mass lives. Fraction  $\leq 1.5$  m: GT 80.7%, matched 89.2%, all 77.3%.

survive, median absolute phantom  $\Delta h$  of 0.00 m.

- **Train performance regresses slightly** (median ride is unaffected; the small TRAIN MAE increase reflects the  $|\vec{a}_h|^2/2g$  horizontal-motion leakage analysed in §12.2). The total error budget on train + test is still substantially better than the baseline.
- **Quality filter still earns its keep.** 0.0% of false positives are rejected by the filter; clean predictions retain 96.8% acceptance.

The boutique pipeline (`src/pipelines/boutique_pipeline.py`) defaults its in-app segmentation to  $|a| - g$  from the same change that produced this section, so the signal the user sees in the Streamlit UI (step 3, "Segmentation") matches what the matched filter actually scores against.

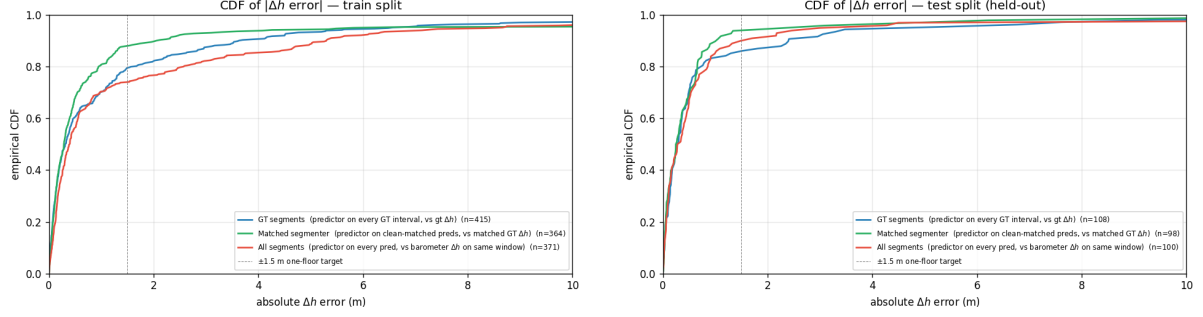


Figure 96: Per-split CDFs. The held-out test split (right) is the key generalisation read: matched-view fraction  $\leq 1.5$  m is 93.9%, all-view fraction  $\leq 1.5$  m is 90.0%.

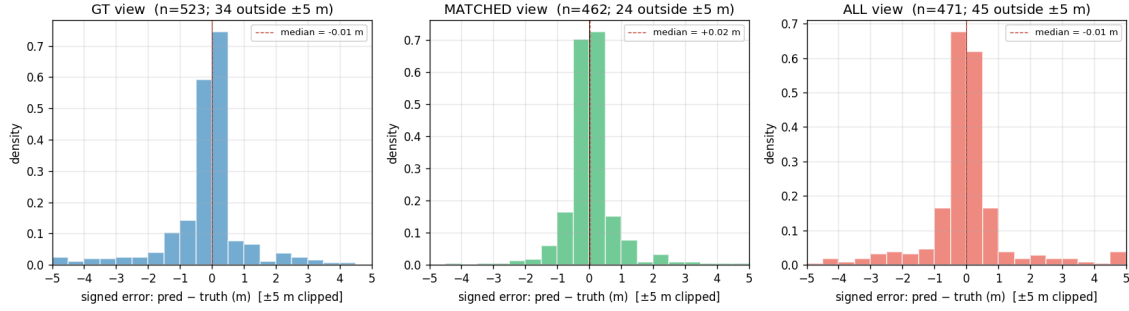


Figure 97: Signed-error PDFs per view. A symmetric, near-zero-mean distribution implies the predictor is not biased toward over-/under-shooting; long tails are where outliers live. Compare visually with the baseline figure in §13 — the  $|a| - g$  tails are markedly thinner.

## 14 Alternative Data

The field campaign documented in §2 is deliberately small and homogeneous: seven elevators across six buildings, all in central Israel, all recorded with the same Android logging app and the same four-phone rig. That is enough to develop and calibrate the pipeline, but it is natural to ask what *external* data exists for elevator rides — both as a check on how unusual our setup is and as a possible source of pre-training or cross-dataset validation. We therefore surveyed publicly described datasets that contain smartphone or wearable *accelerometer* recordings of *labelled elevator rides*; Table 33 collects them. Two columns deserve emphasis because they are the two ingredients our own data is built around: a **barometer** channel (our ground-truth altitude source, §2.2) and an explicit **height-difference / floor-level** label (our prediction target, §4).

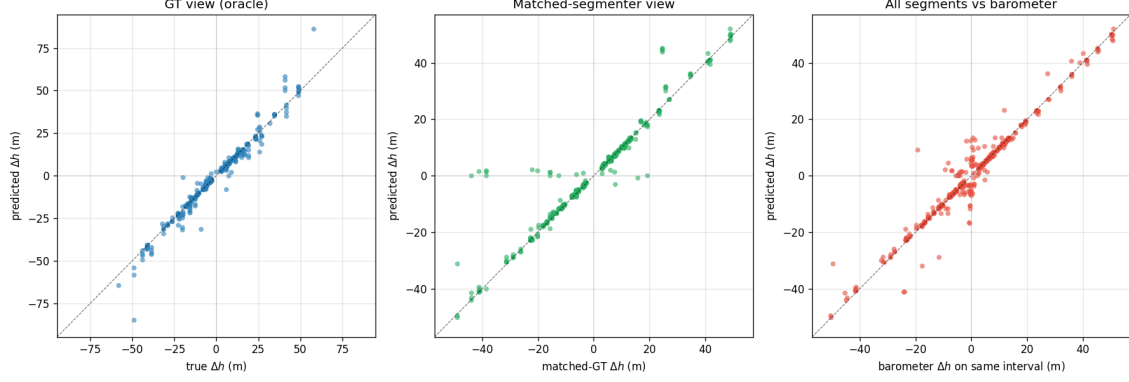


Figure 98: Predicted vs. truth scatter, all three views,  $|a| - g$ . Points off the diagonal are mis-predictions; the cluster width around  $y = x$  reflects the conditional spread of the predictor.

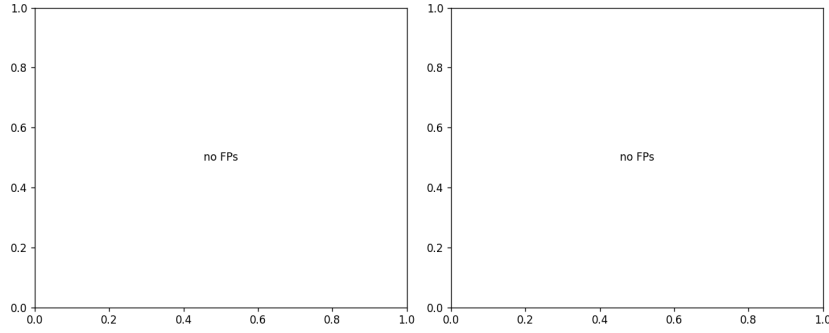


Figure 99: Phantom  $\Delta h$  on false-positive segments. Under  $|a| - g$ , 0 false-positive segments survive the matched-filter pair-filter — the histogram is empty because every segmenter prediction in the corpus cleanly matched a real GT ride. Median signed phantom is +0.00 m, mean absolute phantom is 0.00 m. Compare to the baseline (Section 13) where the FP histogram had non-trivial mass driven by stale- $\hat{g}$ -projection artefacts. The companion “predicted altitude” figure that would normally sit beside this one is omitted: with zero FPs the renderer has nothing to draw.

| Dataset                                                             | Sensors (device)                                                                         | Elevator labelling                                                                    | Baro.          | $\Delta h$ | GT | Public |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------|------------|----|--------|
| USC-HAD<br>(Zhang & Sawchuk, 2012)                                  | Acc, Gyr<br>wearable IMU; 14<br>subj.; 100 Hz                                            | Elevator-up /<br>Elevator-down<br>classes (of 12 ADLs)                                | ×              | ×          | ×  | ✓      |
| ExtraSensory<br>(Vaizman et al., 2017)                              | Acc, Gyr, Mag,<br>audio, GPS, phone-<br>state<br>phone + watch; 60<br>users; in-the-wild | elevator context la-<br>bel (no up/down)                                              | ×              | ×          | ×  | ✓      |
| Indoor-loc. HAR<br>(Zhao et al., <i>Sen-<br/>sors</i> , 2019)       | Acc, Gyr, Mag, <b>Baro</b><br>phone; 10 subj.; 50 Hz;<br>>6 GB                           | up/down elevator,<br>escalator & stairs (9<br>HAR classes; also<br>walk, still, turn) | ✓              | ×          | ×  | †      |
| ELESON<br>(arXiv:2405.03218,<br>2024)                               | Acc, Gyr, Mag<br>phones; 36,420 inst.; 10<br>malls                                       | elevator / escalator<br>/ neither (state, no<br>up/down)                              | ×              | ×          | ×  | ×      |
| IMU floor-height<br>est.<br>(Chen et al., IEEE<br>Sensors J., 2024) | Acc, Gyr, Mag +<br>Baro <sup>‡</sup><br>phone; 50 Hz; i2nav lab                          | elevator + escalator<br>(no stairs)                                                   | ✓ <sup>‡</sup> | ✓          | ✓  | ×      |
| 112                                                                 |                                                                                          |                                                                                       |                |            |    |        |

Table 33: Publicly described datasets containing accelerometer recordings of labelled elevator rides, and how they compare to our campaign on the two properties that matter for this work:

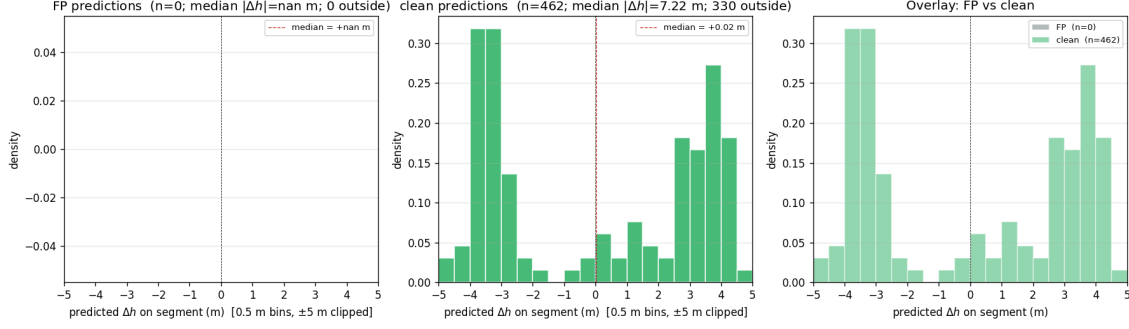


Figure 100: Distribution of predicted  $|\Delta h|$  for FP segments vs. clean (matched-GT) segments. A separable left-right view here means the predictor’s output magnitude alone could gate out FPs; overlap means we depend on the quality filter to do it.

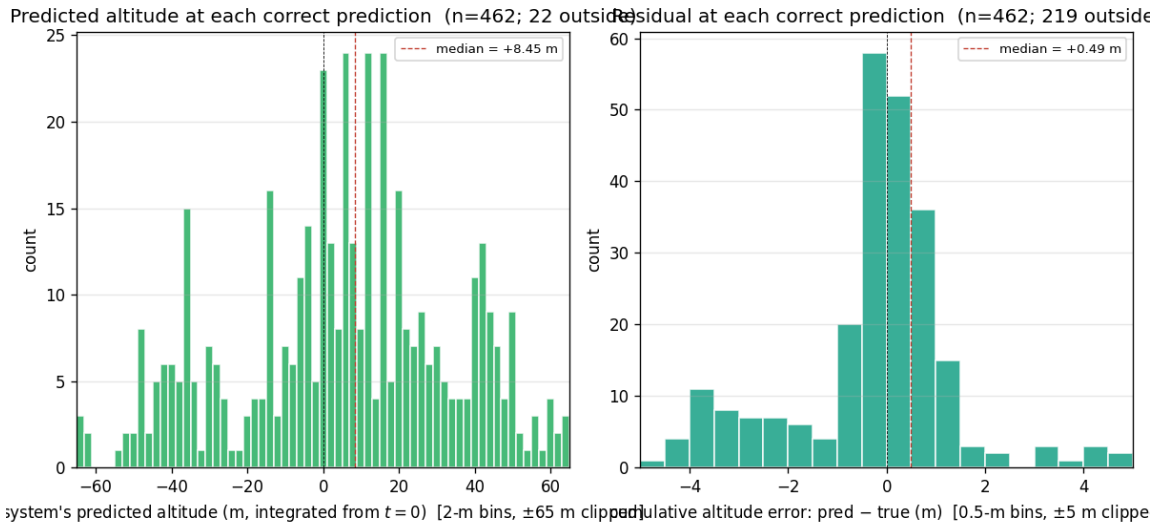


Figure 101: Predicted altitude trajectories for clean (matched) segments. Each line is one experiment; the slope encodes ride direction, the dwell encodes pause-between-rides. Under  $|a| - g$ , lines are visibly straighter (less amplitude collapse from stale- $\hat{g}$  projection).

**Access.** Live entry points at the time of writing: USC-HAD [sipi.usc.edu/had](http://sipi.usc.edu/had); ExtraSensory [extrasensory.ucsd.edu](http://extrasensory.ucsd.edu) (also mirrored on Kaggle); the *Sensors* 2019 indoor-localization paper [PMC6387421](https://pubmed.ncbi.nlm.nih.gov/36387421/); ELESON [arXiv:2405.03218](https://arxiv.org/abs/2405.03218); and the IMU floor-height paper [IEEE Xplore 10648587](https://ieeexplore.ieee.org/abstract/document/10648587).

**What the table shows.** No public dataset we found matches our recording setup on all three axes at once — accelerometer *and* barometer *and* a per-ride height label. Each near match drops one axis: the activity-recognition corpora (USC-HAD, ExtraSensory, ELESON) label elevator use but carry neither a barometer nor a height target; the *Sensors* 2019 indoor-localization set adds a barometer (and even covers stairs) but never ships per-ride  $\Delta h$  and was not, as far as we can tell, ever released; and the IMU height-estimation work of Chen et al. — the one entry that, exactly like our data, carries accelerometer, barometer (as ground truth) *and* a floor-level height all at once — keeps its raw data private. Two large, barometer-rich locomotion datasets — SHL (Sussex-Huawei, 2,800 h, accelerometer + barometer across three phones) and UCI-HAR — are worth noting as generic pre-training material but contain *no* elevator class at all. The empty cell our campaign fills is therefore a barometer-backed, height-labelled, accelerometer recording of real elevator rides.

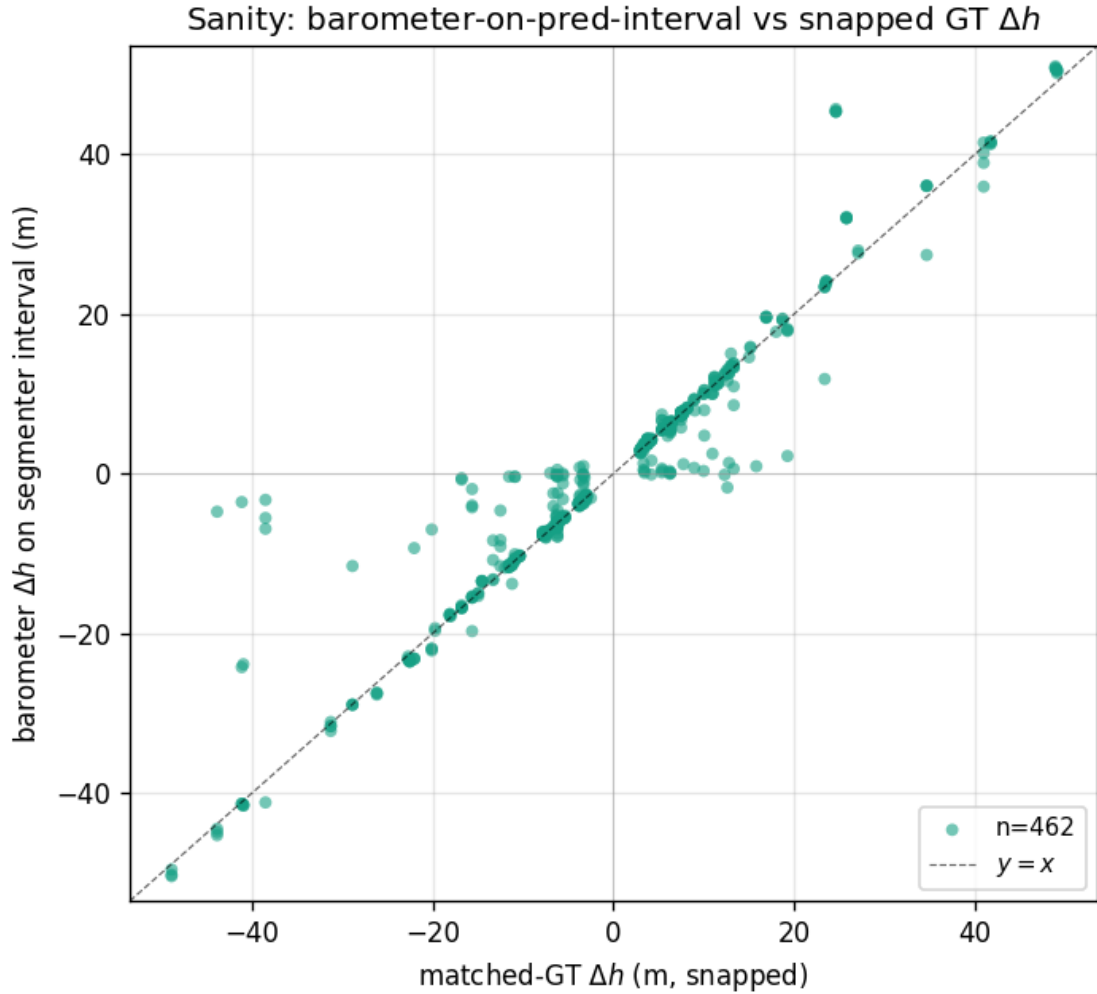


Figure 102: Barometer-derived  $\Delta h$  vs. ground-truth  $\Delta h$  for every matched segment — a sanity check that the "all" view’s barometer truth source agrees with the per-ride GT. Clustering near  $y = x$  validates that the third-view-vs-barometer scoring is honest under the new segmentation as well.

## 15 Gyroscope orientation reconstruction: recovering the true vertical acceleration

The pipeline described so far is accelerometer-only and assumes the phone’s orientation is fixed for the duration of a ride (Sec. 1 notes it carries “no orientation fusion”). This section lifts that assumption. We fuse the phone’s *gyroscope* with its accelerometer to track how the device is oriented *through* the ride, rotate the acceleration into a gravity-aligned world frame, and hand the two stages a cleaner vertical channel  $a_{\text{vert}}$ . We implemented five standard orientation filters, evaluated all of them end-to-end, and adopt the **Mahony** filter, which gave the best segmentation and prediction on our data. The reconstruction is a config-selectable first step of both `Segmenter.detect` and `Predictor.predict` (`reconstruct=none` reproduces the baseline byte-for-byte), so everything below is a drop-in improvement, not a re-architecture.

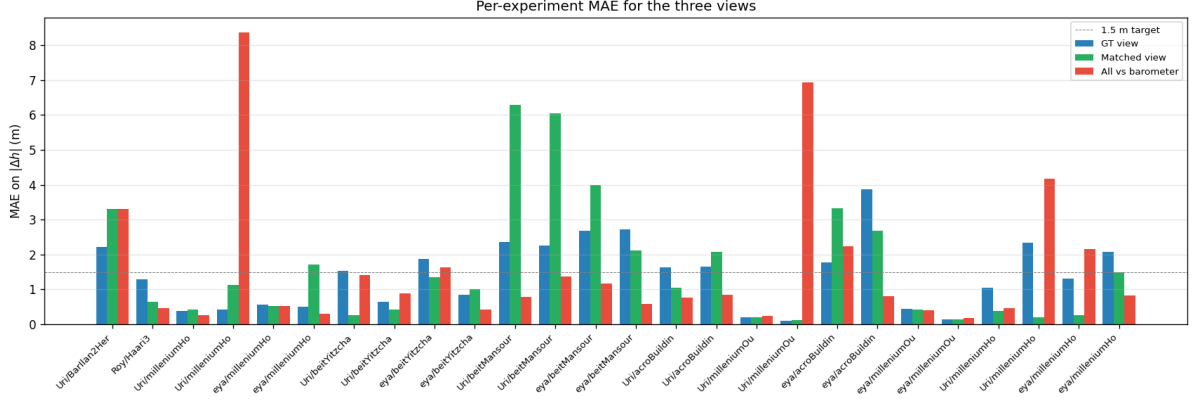


Figure 103: Per-experiment MAE across the 26 experiments, all three views. Bars reveal whether one or two pathological experiments drive the aggregate, vs. uniform performance across phones and buildings.

### 15.1 Why a gyroscope helps

Both stages consume the vertical channel  $a_{\text{vert}}(t) = (\mathbf{a}(t) - \hat{\mathbf{g}}) \cdot \hat{\mathbf{g}} / |\hat{\mathbf{g}}|$ , the accelerometer projected onto a *single* gravity direction  $\hat{\mathbf{g}}$  estimated once from the pre-ride stationary window. At rest the accelerometer reports the specific force  $\mathbf{a} = -\mathbf{g}$ , a vector of magnitude  $g \approx 9.81 \text{ m/s}^2$  pointing to world  $+Z$ ; because the phone is tilted, that vector does not lie along the body  $z$  axis, and its body coordinates encode the tilt (Fig. 109). Projecting onto a fixed  $\hat{\mathbf{g}}$  is therefore exact only while the tilt is frozen.

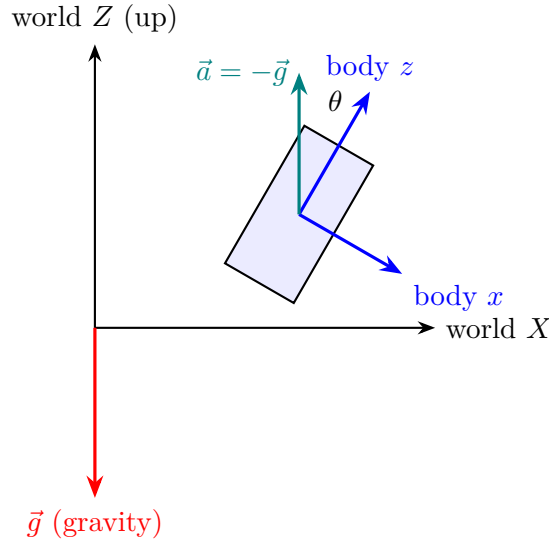


Figure 109: At rest the accelerometer measures the *specific force*  $\vec{a} = -\vec{g}$  (pointing to world  $+Z$ , magnitude  $g$ ). Because the phone is tilted by  $\theta$ , this vector does *not* lie along the body  $z$  axis; its coordinates in the body frame encode the tilt. The vertical channel  $a_{\text{vert}}$  is the projection of  $\vec{a}$  onto the gravity direction, so a wrong tilt estimate contaminates it with a slice of the  $9.81 \text{ m/s}^2$  gravity term.

In practice the phone is held, pocketed, or glanced at, so its tilt drifts over the ride (Fig. 111). A stale  $\hat{\mathbf{g}}$  then leaks a slice of gravity into  $a_{\text{vert}}$ : a tilt error of  $\delta\theta$  injects a spurious acceleration  $g \sin \delta\theta$ , so a mere  $5^\circ$  (0.09 rad) already contributes  $\approx 0.85 \text{ m/s}^2$  — comparable to the ride’s own signal, and, because ZUPT integrates twice, worth metres of  $\Delta h$  error over a 10 s ride.

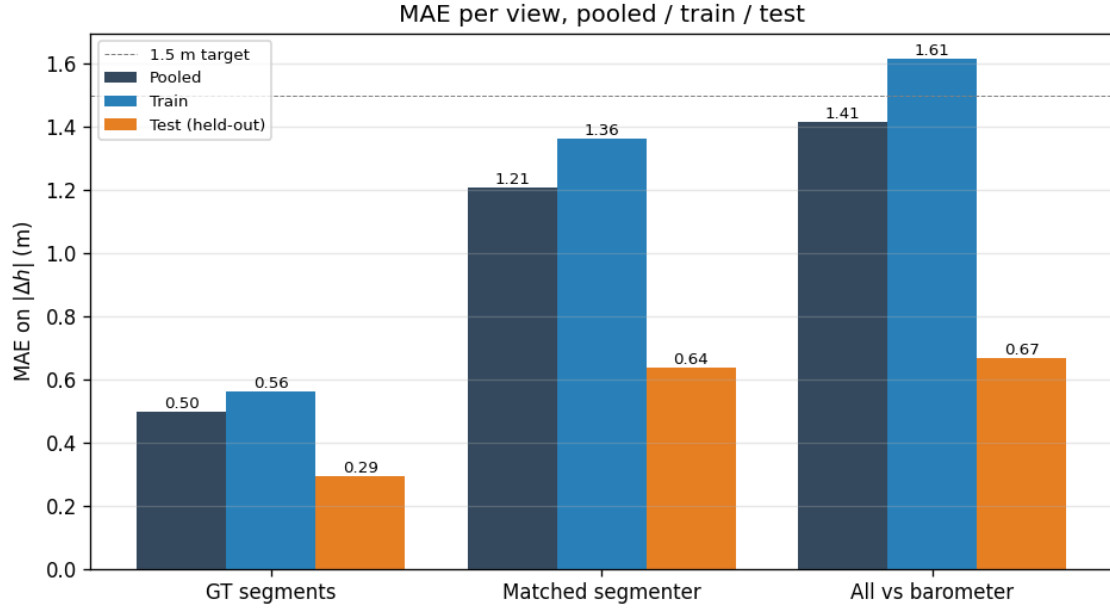


Figure 104: Accepted-only MAE. The quality filter removes a curated set of low-confidence predictions; what remains is the predictor’s operating distribution for downstream consumers that gate on `accepted=True`.

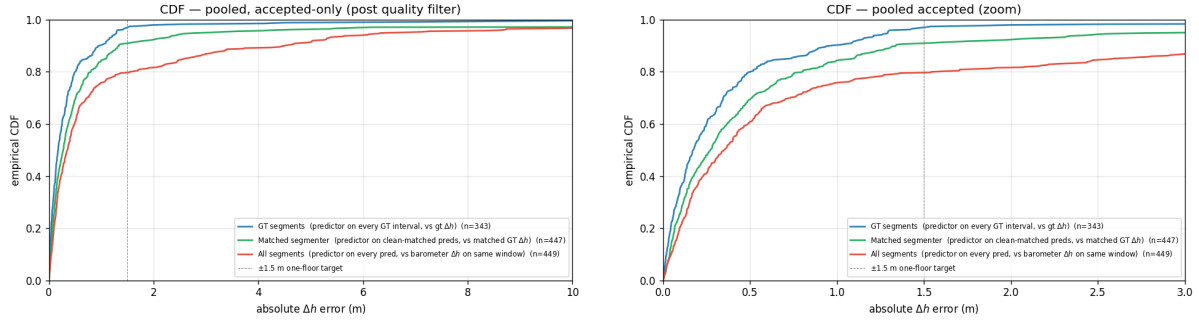


Figure 105: Accepted-only pooled CDF. The right zoom emphasises the  $\leq 1.5$  m operating regime.

The gyroscope measures the phone’s angular rate  $\omega = (\omega_x, \omega_y, \omega_z)$  about its own body axes (Fig. 110). Only *roll* and *pitch*, which tilt the body  $z$  axis away from vertical, affect  $a_{\text{vert}}$ ; *yaw* (rotation about the vertical) leaves the gravity direction unchanged, which is why a magnetometer is unnecessary here. Integrating  $\omega$  tracks how the tilt evolves, so the gravity direction can be followed through the ride instead of assumed constant. The gyroscope alone drifts (its bias integrates without bound); the accelerometer alone is corrupted by the ride’s own motion. Fusing them — gyro for the fast changes, accelerometer as a slow absolute reference whenever  $|\mathbf{a}| \approx g$  — is exactly what an *orientation filter* does.



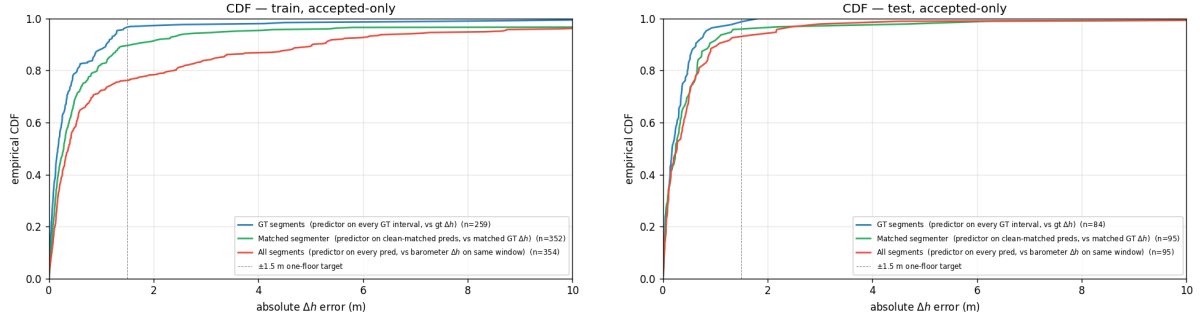


Figure 106: Accepted-only per-split CDFs. Test-split matched-view fraction  $\leq 1.5$  m on accepted predictions only is 95.8%.

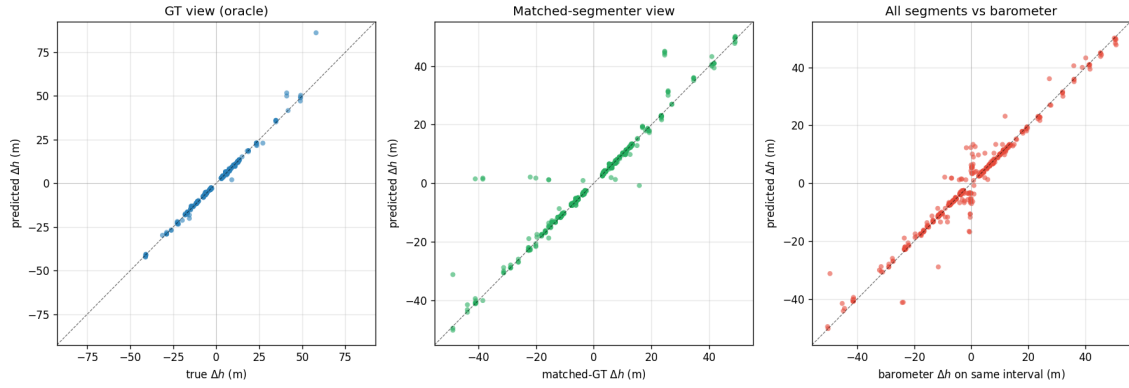


Figure 107: Accepted-only scatter: predicted  $\Delta h$  vs. truth for all three views. Tightness near the diagonal is the deployment story.

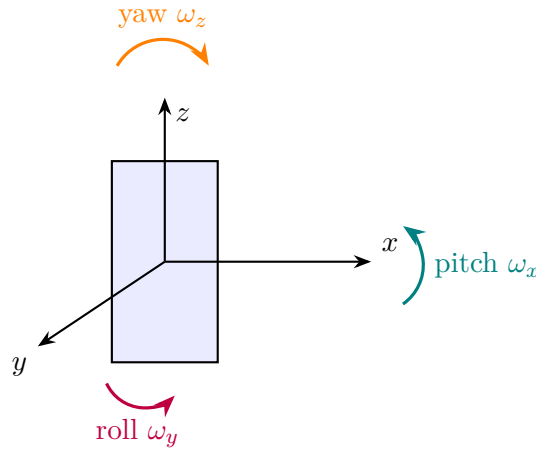


Figure 110: The three body-frame rotations the gyroscope measures as angular rates: *roll* about  $y$ , *pitch* about  $x$ , *yaw* about  $z$ . Only roll and pitch (which together set the *tilt* of  $z$  away from vertical) affect  $a_{\text{vert}}$ ; yaw leaves the gravity direction unchanged and is irrelevant here.

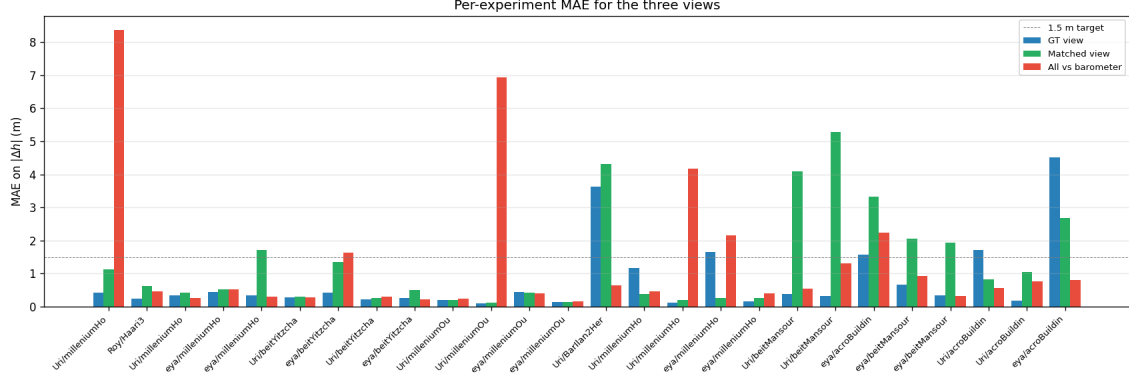


Figure 108: Per-experiment MAE, accepted-only. Compare with the unfiltered Figure 103 to see which experiments contributed rejected predictions.

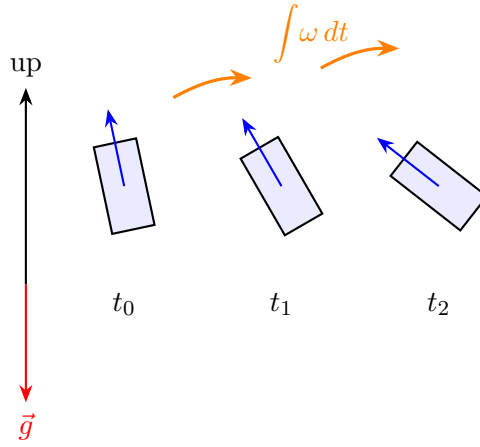


Figure 111: During a ride the phone is handled, so its tilt drifts  $\theta(t_0) \rightarrow \theta(t_1) \rightarrow \theta(t_2)$ . A single gravity vector estimated from the pre-ride window (the baseline) is stale by  $t_2$ , leaking a time-varying bias into  $a_{\text{vert}}$  that mimics a ride pulse. Integrating the gyroscope,  $q(t) = q(t_0) \otimes \int \omega dt$ , tracks the tilt so the projection stays aligned; the accelerometer only re-anchors it when  $|\mathbf{a}| \approx g$ .

## 15.2 Orientation reconstruction pipeline

We represent orientation by a unit quaternion  $q$  mapping the body frame to the world frame (world  $Z = \text{up}$ ), with rotation matrix  $R(q)$ . All five filters share the same skeleton:

1. **Seed.** Average the accelerometer over the opening stationary window and set  $q(t_0)$  to the shortest rotation taking measured gravity onto world  $+Z$  (yaw is left free; it does not affect  $a_{\text{vert}}$ ).
2. **Propagate** with the gyroscope, using the quaternion kinematics

$$\dot{q} = \frac{1}{2} q \otimes \begin{bmatrix} 0 \\ \omega \end{bmatrix}, \quad q_{k+1} = q_k \otimes \exp\left(\frac{1}{2} \omega_k \Delta t\right), \quad (44)$$

where  $\otimes$  is the Hamilton product and  $\exp$  the quaternion exponential of a rotation vector.

3. **Correct** the accumulating gyro drift with the accelerometer: each filter nudges  $q$  so that the predicted gravity direction  $\hat{v} = R(q)^\top \hat{z}$  aligns with the measured one  $\hat{a} = \mathbf{a}/|\mathbf{a}|$ . The correction is down-weighted by the *accel-trust* gate

$$w = \exp\left(-\frac{1}{2}((|\mathbf{a}| - g)/(\tau g))^2\right) \in (0, 1], \quad (45)$$

which collapses to 0 during the ride’s acceleration pulses (when  $|\mathbf{a}| \neq g$ ) so ride motion is never mistaken for tilt.

4. **Rotate and re-emit.** With the tracked  $q(t)$ ,  $\mathbf{a}_w = R(q) \mathbf{a}$  and  $a_{\text{vert}} = \mathbf{a}_{w,z} - g$ . To leave the stages’ internal gravity estimator untouched we feed them the *virtually-flat-phone* signal  $(\mathbf{a}_{w,x}, \mathbf{a}_{w,y}, \mathbf{a}_{w,z} + g)$  — gravity restored on  $+z$  — so their  $|\hat{\mathbf{g}}| \approx g$  gate still fires and the projection returns the true signed vertical acceleration.

### 15.3 The chosen filter: Mahony

We adopt the **Mahony** passive complementary filter on  $SO(3)$ . It won both stages on our data (Sec. 15.5) while being the cheapest to run and one of only two candidates (with the ESKF) that estimate the gyro bias online. Mahony forms the attitude error directly as the vector cross product between the measured and predicted gravity directions,

$$\mathbf{e} = \hat{\mathbf{a}} \times \hat{\mathbf{v}}, \quad \hat{\mathbf{v}} = R(q)^\top \hat{\mathbf{z}}, \quad (46)$$

so  $\mathbf{e}$  points along the axis that rotates the estimate onto the measurement and  $\|\mathbf{e}\| \approx \sin(\text{tilt error})$ . It then drives a *proportional–integral* correction back into the gyro kinematics:

$$\dot{\mathbf{b}} = -k_I w \mathbf{e} \quad (\text{bias integrator}), \quad (47)$$

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} - \mathbf{b} + k_P w \mathbf{e} \end{bmatrix} \quad (\text{corrected kinematics}). \quad (48)$$

The proportional term  $k_P w \mathbf{e}$  pulls the fast gyro estimate toward the accelerometer’s gravity reference; the integral term  $-k_I w \mathbf{e}$  learns the slow gyro bias  $\mathbf{b}$  and subtracts it, so the estimate does not drift during the long, low-excitation cruise phase of a ride. The trust weight  $w$  from Eq. (45) multiplies *both* corrections, freezing the update to pure gyro integration exactly when the elevator is accelerating. Discretising with a renormalisation each sample,

$$\mathbf{b}_{k+1} = \mathbf{b}_k - k_I w_k \mathbf{e}_k \Delta t, \quad \mathbf{q}_{k+1} = \frac{\mathbf{q}_k \otimes \exp(\frac{1}{2}(\boldsymbol{\omega}_k - \mathbf{b}_{k+1} + k_P w_k \mathbf{e}_k) \Delta t)}{\|\mathbf{q}_k \otimes \exp(\cdot)\|}. \quad (49)$$

Each update is a cross product and one quaternion product ( $O(1)$  per sample), has two interpretable gains  $(k_P, k_I)$ , and degrades gracefully: with  $k_I = 0$  it reduces to a plain complementary filter.

### 15.4 The other four filters and how they differ

All four solve the same fuse-gyro-with-gravity problem; they differ in *how* the accelerometer correction is formed.

**Complementary.** The minimal baseline: integrate the gyro (Eq. 44), then apply a fraction  $\alpha w$  of the shortest-arc rotation  $\delta \mathbf{q}$  taking  $R(q)\hat{\mathbf{a}}$  onto  $\hat{\mathbf{z}}$ ,  $\mathbf{q} \leftarrow \text{slerp}(1, \delta \mathbf{q}; \alpha w) \otimes \mathbf{q}$ . No bias state, no integral term — a single blend gain  $\alpha$ . Mahony *is* this filter plus a bias integrator expressed in the Lie algebra.

**Madgwick.** Replaces the cross-product error with one step of *gradient descent* on the gravity-alignment cost  $f(q) = R(q)^\top \hat{\mathbf{z}} - \hat{\mathbf{a}}$ :  $\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \boldsymbol{\omega} - \beta w \nabla f / \|\nabla f\|$  with  $\nabla f = J^\top f$ . A single tunable rate  $\beta$ . Equivalent in spirit to Mahony’s proportional term but with a normalised-gradient step; at our default  $\beta$  it over-corrected on some rides and *hurt* segmentation (Sec. 15.5).

**Valenti / AQUA.** Computes the accelerometer’s implied orientation *algebraically* (a closed-form quaternion  $\Delta q_{acc}$  rotating  $R(q)\hat{a}$  onto  $\hat{z}$ ) and fuses it with the gyro prediction through an *adaptive* gain: the interpolation factor ramps linearly to zero once  $\|\mathbf{a}\| - g/g$  exceeds a threshold, and switches LERP→SLERP for large corrections. More principled gain scheduling than a fixed  $\alpha$ , but no bias state.

**ESKF.** The probabilistic optimum: an *error-state Kalman filter* carrying a covariance  $P$  over the small attitude error  $\delta\theta$  and bias error  $\delta b$ . The gyro propagates the nominal state and  $P$ ; the accelerometer enters as a measurement with Jacobian  $H = [[\hat{v}]_{\times} \ 0]$  and noise  $R = (\sigma_a/w)^2 I$  (the trust gate enters as measurement noise). It computes a Kalman gain, injects  $q \leftarrow q \otimes [1, \frac{1}{2}\delta\theta]$ , and resets. Most accurate in principle — and best on the tightest-CI prediction metric — but the heaviest to run and to tune (needs process/measurement noise priors).

Mahony sits at the sweet spot: it keeps the ESKF’s online bias estimation and the complementary filter’s simple cost, wins our two end-to-end metrics, and needs no noise model.

## 15.5 Results: how much reconstruction improves each stage

The tables and figures here are produced by `scripts/reconstruction_comparison_report.py` (all six variants through both stages, `--kind all`) and by the Mahony configuration of `src.prediction.evaluation.evaluateOnData / src.segmentation.evaluate.evaluateOnData`. All numbers are on the full resolved set (34 experiments; 619 prediction segments, 627 ground-truth rides).

**Prediction.** Table 34 shows reconstruction roughly *halves* the mean absolute  $\Delta h$  error for both predictors — ZUPT  $2.55 \rightarrow 1.07$  m, trapezoid  $1.73 \rightarrow 0.94$  m — and shrinks RMSE far more (ZUPT  $18.8 \rightarrow 3.2$  m), because the worst damage was on a handful of heavily rotated rides whose fixed-gravity error ran to tens of metres. The fraction of segments within one floor (1.5 m) rises (ZUPT  $75.8 \rightarrow 86.9\%$ ) and the median CI narrows, i.e. the predictions get both more accurate and more confident.

| Algorithm | Reconstruction  | MAE (m)      | RMSE (m)     | Cov@90       | $\leq 1.5$ m | med. CI (m)  |
|-----------|-----------------|--------------|--------------|--------------|--------------|--------------|
| ZUPT      | none (baseline) | 2.545        | 18.827       | 91.9%        | 75.8%        | 2.256        |
|           | <b>Mahony</b>   | <b>1.068</b> | <b>3.238</b> | 91.6%        | <b>86.9%</b> | <b>1.237</b> |
| Trapezoid | none (baseline) | 1.734        | 11.313       | 83.5%        | 84.8%        | 0.894        |
|           | <b>Mahony</b>   | <b>0.943</b> | <b>2.734</b> | <b>86.4%</b> | <b>88.0%</b> | <b>0.892</b> |

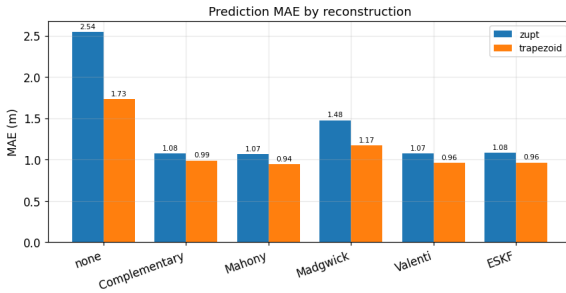
Table 34: Prediction accuracy on the full set, baseline (no reconstruction) vs. Mahony. Reconstruction halves MAE and cuts RMSE by 3–6 $\times$ .

**Segmentation.** Table 35 shows the detector catches more rides ( $498 \rightarrow 550$  of 627) while its false positives collapse ( $43 \rightarrow 8$ ), lifting  $F1^*$  from 0.847 to 0.921 and IoU-F1 from 0.726 to 0.774: a cleaner  $a_{vert}$  makes the trapezoid template match where it should and stop matching tilt artefacts where it should not.

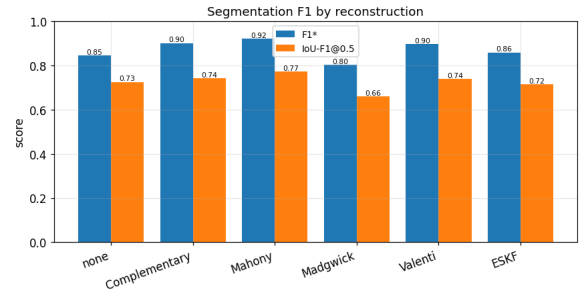
| Reconstruction  | $n_{\text{gt}}$ | detected   | missed    | FP       | $F1^*$       | IoU-F1@0.5   | precision    |
|-----------------|-----------------|------------|-----------|----------|--------------|--------------|--------------|
| none (baseline) | 627             | 498        | 116       | 43       | 0.847        | 0.726        | 0.907        |
| <b>Mahony</b>   | 627             | <b>550</b> | <b>67</b> | <b>8</b> | <b>0.921</b> | <b>0.774</b> | <b>0.970</b> |

Table 35: Segmentation accuracy on the full set, baseline vs. Mahony. False positives drop  $43 \rightarrow 8$ ; the detector both finds more rides and fires spuriously far less often.

**Which filter, and an honest caveat.** Across all five filters Mahony was the best or tied-best on every end-to-end metric except the very tightest CIs, where the heavier ESKF edges ahead. Not every filter helps: at its default gain **Madgwick over-corrects and degrades segmentation**, driving false positives up to 92 (worse than the baseline’s 43) — a reminder that an orientation filter must be validated on the downstream task, not just on attitude error. Figure 112 summarises both stages across all variants, and Figure 113 shows the mechanism on one rotated ride: the fixed-gravity  $a_{\text{vert}}$  carries a tilt-induced bias in the calm mid-ride section that Mahony removes.



(a) Prediction MAE by reconstruction.



(b) Segmentation  $F1^*$  / IoU-F1 by reconstruction.

Figure 112: Every reconstruction beats the baseline on prediction MAE; Mahony is best or tied-best on both stages, while Madgwick’s segmentation regresses.

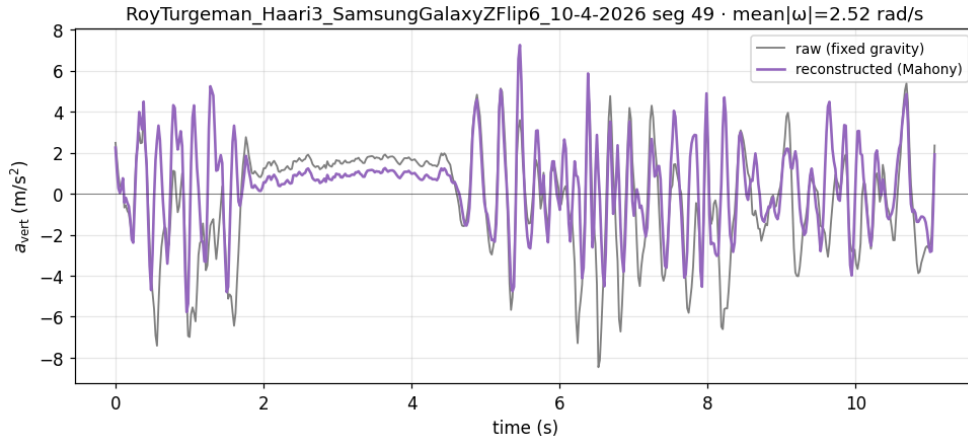


Figure 113: Raw fixed-gravity  $a_{\text{vert}}$  vs. Mahony-reconstructed vertical acceleration on a rotating ride (mean  $|\omega| \approx 2.5$  rad/s). The raw trace carries a tilt-induced bias in the calm mid-ride section that reconstruction removes.

The benefit is concentrated exactly where the model predicts: Figure 114 plots each ride’s rotation energy (mean  $|\omega|$ ) against the error reduction reconstruction buys — near-zero for upright rides, large for the most-rotated ones.

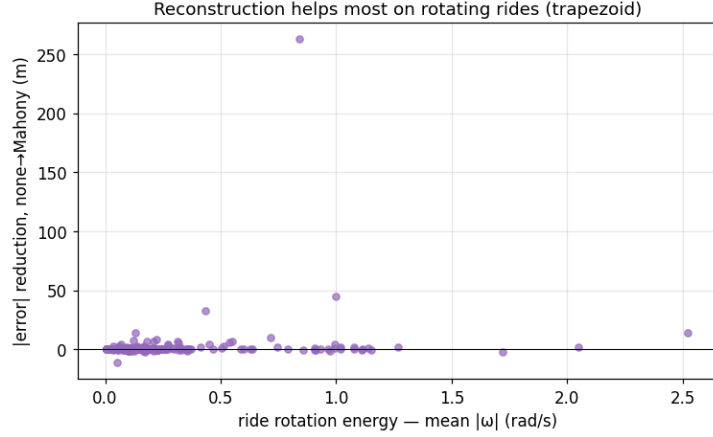


Figure 114: Per-ride rotation energy vs. the trapezoid  $|\text{error}|$  reduction from Mahony reconstruction: the gain concentrates on the most-rotated rides, and is negligible on upright ones (so reconstruction never hurts the easy case).

**Calibration holds out.** Because the trapezoid/ZUPT confidence intervals are conformally calibrated, we refit the conformal multiplier on the *train* split with Mahony reconstruction and scored the held-out *test* split with it (Table 36). Test-split rides are largely upright, so the gain is smaller than on the full set; even so, Mahony *improves* the within-one-floor rate on both predictors (ZUPT  $87.0 \rightarrow 91.7\%$ , trapezoid  $90.7 \rightarrow 93.5\%$ ) and tightens or matches the median CI, all while its 90% interval keeps  $\geq 90\%$  coverage. Reconstruction does no harm on upright rides and helps even here, and it does not break calibration.

| Algorithm | Reconstruction  | $n_{\text{clean}}$ | Cov@90 (GT-clean) | $ \tilde{e} $ (m) | $\tilde{w}$ (m) | $\leq 1.5$ m |
|-----------|-----------------|--------------------|-------------------|-------------------|-----------------|--------------|
| ZUPT      | none (baseline) | 108                | 94.4%             | 0.19              | 1.00            | 87.0%        |
|           | <b>Mahony</b>   | 108                | 95.4%             | 0.21              | <b>0.91</b>     | <b>91.7%</b> |
| Trapezoid | none (baseline) | 108                | 91.7%             | 0.29              | 0.83            | 90.7%        |
|           | <b>Mahony</b>   | 108                | 93.5%             | <b>0.27</b>       | 0.84            | <b>93.5%</b> |

Table 36: Mahony with its own conformal calibration (refit on train, scored on the held-out test split). Coverage of the 90% CI is maintained ( $\geq 90\%$ ) while the within-one-floor rate improves on both predictors, even on this mostly-upright split.

## 16 Walkthrough: the Mahony filter step by step

This section traces one run of the Mahony filter (§15.3) end to end: the two sensor streams that enter, the orientation and bias that leave, and the fixed operation sequence each sample runs. Figure 115 is the per-sample dataflow; Figure 116 shows every internal quantity on a controlled rotating ride.

**What we do.** We turn a tilted, rotating phone’s raw inertial stream into the real vertical acceleration  $a_z$ . The phone’s own  $z$ -axis is not world-vertical, so we track the device orientation  $q(t)$  every sample, rotate the measured acceleration into the world frame, and subtract gravity. The gyroscope tracks orientation change accurately but drifts; the accelerometer fixes the gravity direction over the long term but is corrupted by the ride’s own motion. Mahony fuses the two with a proportional–integral correction on  $SO(3)$ .

**Input.** Two body-frame streams, resampled onto the accelerometer clock at rate  $f_s$  (§15):

- accelerometer  $\mathbf{a}_k \in \mathbb{R}^3$  (m/s<sup>2</sup>) — specific force. At rest it reads  $+g$  along the phone’s sensed up; it carries gravity plus the ride’s linear acceleration.
- gyroscope  $\omega_k \in \mathbb{R}^3$  (rad/s) — angular rate about the phone’s own axes.

Two scalars seed the run: the initial orientation  $q_0$  from the opening stationary window, and the gravity magnitude  $g$ .

**Output.** The per-sample body→world orientation  $q_k$  and gyro-bias estimate  $b_k$ . Rotating  $\mathbf{a}_k$  by  $q_k$  and removing gravity yields  $a_z$ , the signed vertical acceleration the height predictors consume.

**The iteration.** The filter is a single loop over samples. It carries two state variables between iterations — the orientation  $q$  and the bias estimate  $b$  — and updates both each sample through the fixed sequence of Figure 115:

1. **Predict** where gravity should point from the current orientation:  $\hat{v} = R(q_k)^\top \hat{z}$ , the world up-axis in body coordinates.
2. **Measure** where gravity actually points:  $\hat{a} = \mathbf{a}_k / |\mathbf{a}_k|$ .
3. **Form the error** as a cross product  $e = (\hat{a} \times \hat{v}) w$  (Eq. 46). Its axis is the rotation aligning prediction with measurement and  $\|e\| \approx \sin(\text{tilt error})$ . The trust gate  $w$  (Eq. 45) scales it, so the correction vanishes when  $|\mathbf{a}| \neq g$ .
4. **Update the bias** by integrating the error:  $b_{k+1} = b_k - k_I e \Delta t$  (Eq. 47).
5. **Correct the rate:**  $\omega' = \omega_k - b_{k+1} + k_P e$  (Eq. 48) — subtract the learned bias, then steer toward gravity by  $k_P e$ .
6. **Integrate** the corrected rate into the orientation:  $q_{k+1} = q_k \otimes \exp(\frac{1}{2} \omega' \Delta t)$ , renormalised (Eq. 49).

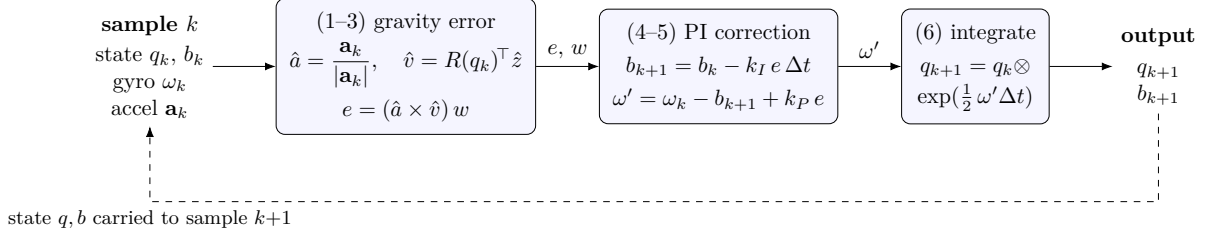


Figure 115: One iteration of the Mahony loop. Two state variables — orientation  $q$  and bias  $b$  — persist across samples (dashed feedback). Each sample predicts gravity from  $q$ , compares it with the accelerometer to form the gated error  $e$ , integrates  $e$  into the bias, corrects the gyro rate, and integrates that back into  $q$ .

**What each iteration’s output means.** Every sample emits an updated  $(q_{k+1}, b_{k+1})$ . The orientation  $q_{k+1}$  is the phone’s current attitude: panel (5) of Figure 116 shows the tilt it encodes tracking the true tilt to within a degree. The bias  $b_{k+1}$  is the running estimate of the gyro’s constant offset; panel (4) shows all three axes climbing to the injected true bias (dashed) within a few seconds, after which the gyro stays de-biased for the rest of the ride. The intermediate error  $e$  and gate  $w$  explain how. Panel (2) is the gravity-direction error the correction nulls; panel (3) shows  $w$  collapsing during the two acceleration pulses (shaded), which freezes the accelerometer correction exactly when ride motion would otherwise read as tilt. Panel (6) is the payoff: Mahony’s  $a_z$  recovers the true trapezoid, while the fixed-frame projection, which ignores the  $30^\circ$  rotation, leaks gravity and distorts both pulses.



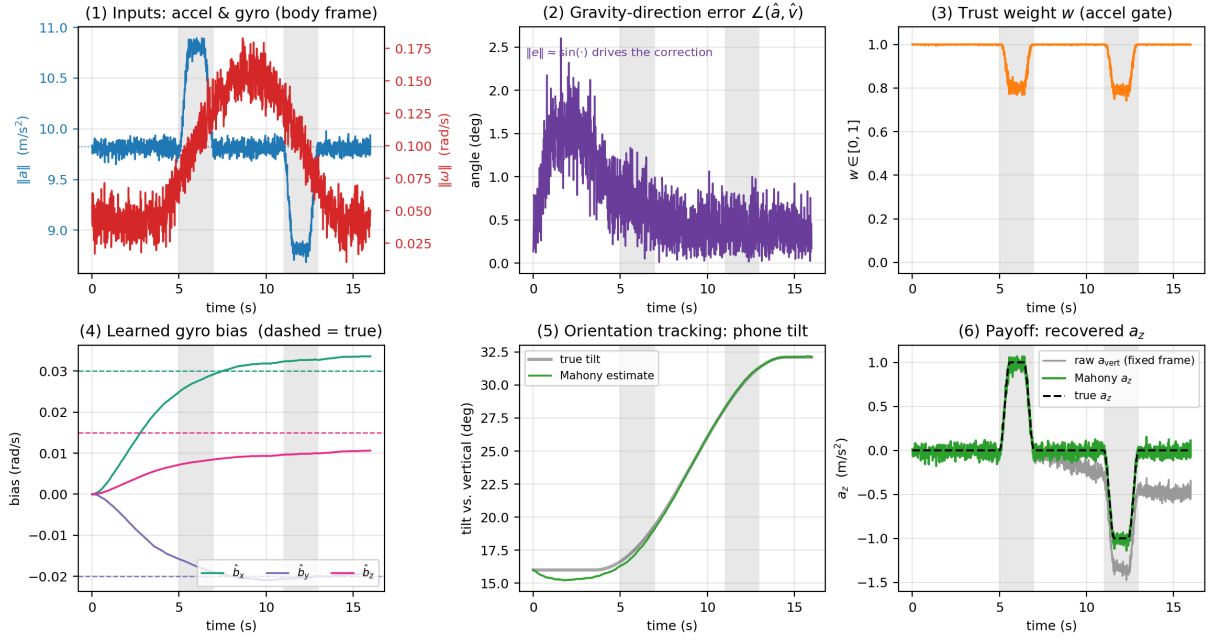


Figure 116: **The Mahony filter on a controlled rotating up-ride**, generated by `scripts/make_mahony_walkthrough.py`. A phone at  $16^\circ$  tilt rotates smoothly to  $32^\circ$  while riding an elevator whose true vertical acceleration is a two-lobe trapezoid (accelerate, cruise, decelerate); we inject a constant gyro bias. **(1)** the two inputs:  $\|\mathbf{a}\|$  leaves  $g$  only inside the pulses (shaded),  $\|\omega\|$  rises with the hand rotation. **(2)** the angle between measured and predicted gravity, whose sine is  $\|e\|$ . **(3)** the trust gate  $w$ , dropping to  $\approx 0.8$  in each pulse. **(4)** the learned bias converging to the injected truth (dashed). **(5)** the estimated tilt tracking the true tilt. **(6)** the recovered  $a_z$ : Mahony (green) matches the truth (dashed); the fixed-frame baseline (grey) does not. The shipped `MahonyReconstructor` produces every curve, and the script asserts its instrumented trace matches bit-for-bit.